

Gyanaranjan Moharana

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SUMMARY

An enthusiastic and motivated student with a strong foundation in Python programming, Machine Learning, and AI fundamentals. Skilled in building and experimenting with ML models, data preprocessing, and problem-solving using Python libraries. Interested in applying technical knowledge in real-world projects through an internship opportunity, while continuously learning and contributing to innovative AI-driven solutions.

WORK EXPERIENCE

IoT Engineer

Aug 2025 - Oct 2025

Worked on the design and development of a solar-powered Smart Aeration System for water quality management. Integrated ESP32 micro controller, SIM800L GSM module, and sensors (Dissolved Oxygen, pH, Turbidity) to monitor real-time water parameters. Implemented IoT-based communication for remote data logging and control. Designed an aerodynamic paddle-based aerator to optimize oxygenation efficiency. Focused on sustainable energy use, automation, and improving aquaculture productivity.

Software Developer

- Acted as a core developer focusing on problem-solving and feature implementation for both Ed-tech and Fleet management software.
- Developed and maintained scalable web-based applications for Ed-tech platforms (handling student learning management and online assessments for [Number]+ users) and Fleet Management systems (optimizing routes and real-time monitoring).
- Worked on backend API development, optimized database design, and integrated machine learning automation tools, improving overall system efficiency and reducing processing time.

PROJECTS

Voice-Guided Object Detection System: Built a navigation assistant using YOLOv8, mathematical depth estimation, and multithreading for real-time obstacle detection and audio feedback, achieving 20 FPS processing.

Disaster Detection AI System: Developed an AI-driven model leveraging [Insert Tech Stack, e.g., CNNs/IoT sensors] to identify and map disaster zones, improving rapid response analysis by 92%.

Student Attendance Management System (SAMS): Developed a comprehensive tracking system featuring a React (Vite) frontend and a Python/FastAPI backend, automating database records for [Number] students.

Smart Aquaculture System: Built using ESP32 and IoT sensors to automate water supply and monitoring, reducing manual intervention by 70%.

Fleet Management System: Real-time fleet tracking with an optimized database backend to support high-frequency location updates.

Ed-Tech Platform: Scalable e-learning system with advanced analytics and secure authentication features.

Face Recognition System: CNN-based real-time face detection and recognition model achieving 89% accuracy.

Placement Portal Web App: Web-based application for managing student placements and tracking opportunities efficiently.

EDUCATION

2023 - Present	Bachelor of Science (BS) Degree in Data Science and Applications	IITM (CGPA: 7.13)
	<i>Relevant Coursework: Data Mining Algorithms, Deep Learning Architectures</i>	
2023 - 2027	Bachelor's Degree in Information Technology at Odisha University Of Technology and Research	(CGPA: 8.3)
	<i>Relevant Coursework: Computer Vision, Software Engineering, Database Management</i>	
2022	Class 12th From CHSE Board, Odisha	(82%)
2020	Class 10th HSE Board, Odisha	(83.67%)

CERTIFICATIONS

Certification Of Appreciation	[Introduction to Deploy ML Apps at Lightning Speed using GCP]	2025
Certification Of Participation	[Entrepreneurship-cum-skill development programme in AI and ML]	(8/Sep - 19/Nov)2025

SKILLS

Python	Strong programming foundation with experience in data analysis and model development using NumPy, Pandas, and Matplotlib
Machine Learning	Neural Networks, Backpropagation, LSTM, Image Segmentation, Object Detection
Machine Learning Tools	Scikit-learn, TensorFlow, YOLOv8
Database Management	PostgreSQL, MySQL, SQL Queries, Joins, Database Design
IoT	ESP32, SIM800L GSM Module, Sensor Integration (DO, pH, Turbidity), Embedded Systems
Web Technologies	HTML, CSS, JavaScript, React (Vite), FastAPI
Soft Skills	Communication, Teamwork, Analytical Thinking, Problem-Solving